

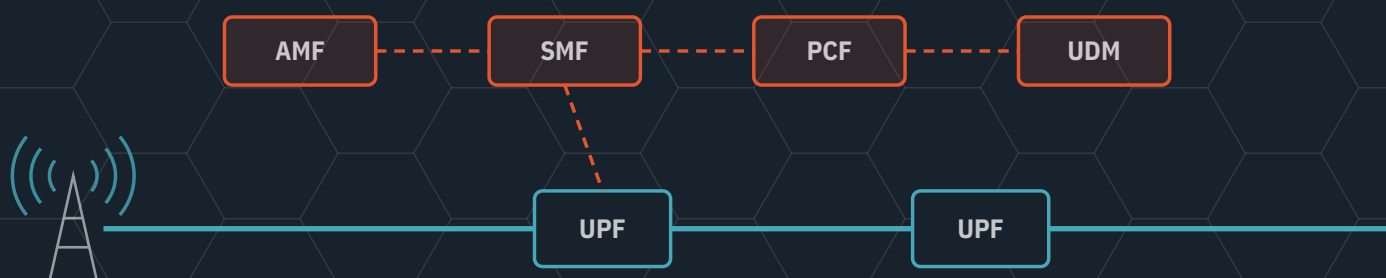
Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 19

5G Core Architecture

معماری هسته 5G



Lecture Notes

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1 5G Core Overview

The 5G Core (5GC) network is defined in **3GPP TS 23.501** and represents a fundamental redesign of the mobile core network. Unlike the 4G EPC which used point-to-point interfaces between monolithic network elements, the 5GC adopts a **Service-Based Architecture (SBA)** where Network Functions (NFs) expose services via RESTful APIs over HTTP/2.

Key Design Principles

- **Service-Based Architecture (SBA):** NFs communicate via service-based interfaces (SBI) using HTTP/2 and JSON
- **Control and User Plane Separation (CUPS):** Complete decoupling of CP and UP
- **Stateless NF Design:** State stored externally (UDSF) enabling horizontal scaling
- **Network Slicing Native:** Built-in support for multiple logical networks on shared infrastructure
- **Cloud-Native:** Designed for containerized, microservice-based deployment
- **Registration-Based NF Discovery:** NFs register with NRF and discover peers dynamically

2 Reference Architecture

The 5GC architecture has two equivalent representations defined in TS 23.501:

2.1 Service-Based Interface (SBI) Representation

In this view, all control-plane NFs connect to a common service bus and expose/consume services.

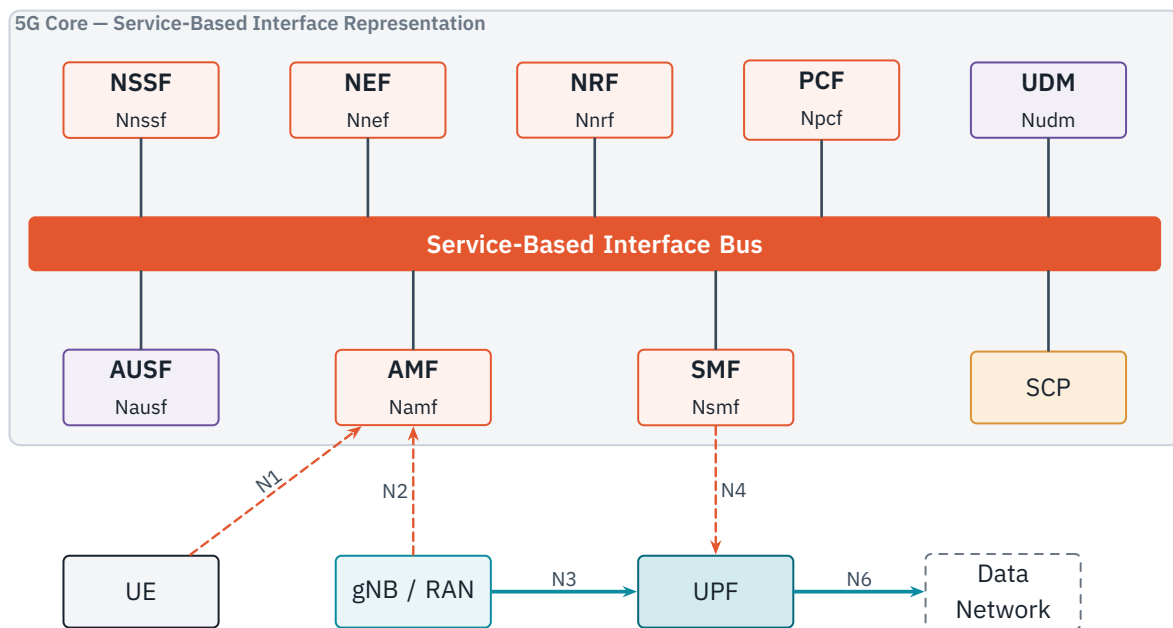


Figure 19.1 5G core in the service-based representation

2.2 Reference Point Representation

In this view, explicit point-to-point reference points are shown between NFs.

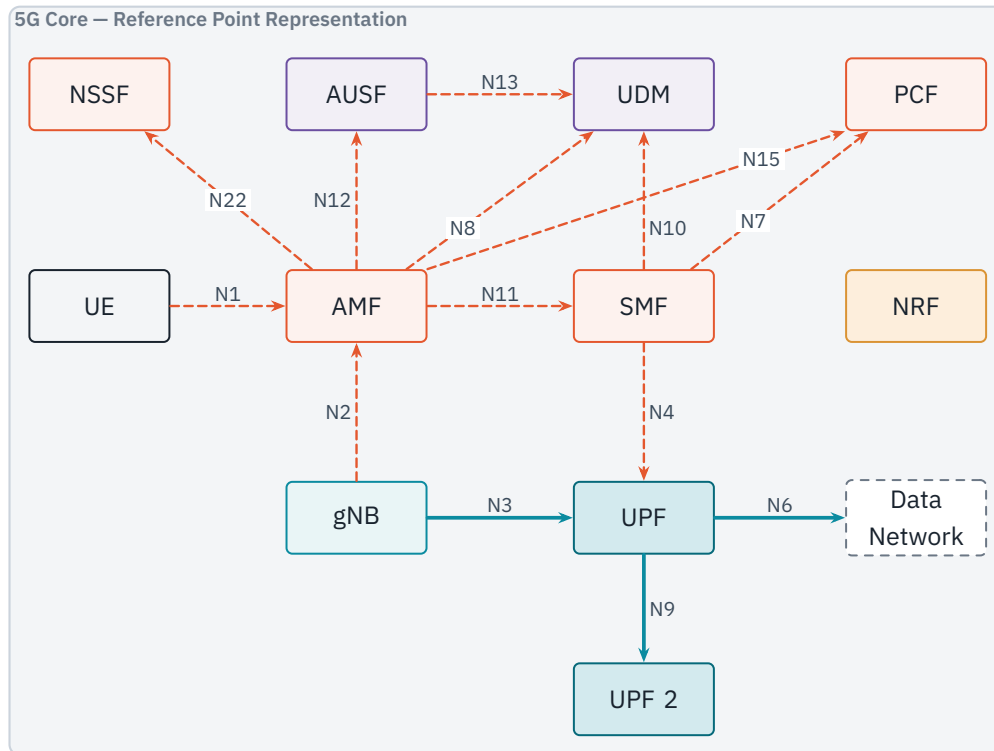


Figure 19.2 5G core in the reference-point representation

3 Network Functions in Detail

3.1 Access and Mobility Management Function (AMF)

The AMF is the single point of contact for all UE-related signaling from the RAN. It terminates the N1 (NAS) and N2 (NGAP) interfaces.

Key Responsibilities:

- **NAS Termination:** All NAS messages (Registration, Service Request, PDU Session Establishment) terminate at AMF
- **Registration Management:** Handles UE registration, deregistration, and periodic registration updates
- **Connection Management:** Manages RRC signaling context via N2 toward gNB
- **Mobility Management:** Handover signaling, tracking area management, paging coordination
- **Security:** Initiates authentication (delegates to AUSF), manages NAS security context (ciphering + integrity)
- **Slice Selection:** Interacts with NSSF to determine the appropriate network slice

Key Identifiers:

Identifier	Description
GUAMI	Globally Unique AMF Identifier = MCC + MNC + AMF Region ID + AMF Set ID + AMF Pointer
5G-GUTI	5G Globally Unique Temporary Identifier = GUAMI + 5G-TMSI (assigned to UE by AMF)
AMF Region ID	8 bits — identifies a region within a PLMN
AMF Set ID	10 bits — identifies an AMF Set within a region
AMF Pointer	6 bits — identifies a specific AMF within a set

Interfaces:

- **N1:** UE ↔ AMF (NAS signaling, carried transparently over RAN)
- **N2:** RAN ↔ AMF (NGAP protocol over SCTP)
- **N8:** AMF ↔ UDM (subscription data retrieval)
- **N11:** AMF ↔ SMF (session management messages)
- **N12:** AMF ↔ AUSF (authentication procedures)
- **N14:** AMF ↔ AMF (UE mobility between AMFs)
- **N15:** AMF ↔ PCF (access and mobility policy)
- **N22:** AMF ↔ NSSF (slice selection)

Registration Area:

- Composed of one or more Tracking Areas (TAs)
- UE only performs Registration Update when moving outside its Registration Area
- Larger registration areas → less signaling, more paging load
- AMF allocates the Registration Area based on UE mobility pattern

3.2 Session Management Function (SMF)

The SMF manages PDU sessions end-to-end. It selects and controls the UPF(s) that form the user plane path.

Key Responsibilities:

- **PDU Session Management:** Establishment, modification, and release of PDU sessions
- **UPF Selection and Control:** Selects appropriate UPF(s), programs forwarding rules via N4/PCF
- **IP Address Allocation:** Assigns IPv4/IPv6 addresses to UEs (or delegates to external DHCP)
- **QoS Enforcement:** Installs QoS rules in UPF based on PCF policies
- **Roaming:** Handles home-routed and local breakout scenarios
- **Downlink Data Notification:** Triggers paging via AMF when downlink data arrives for idle UE

PDU Session Types:

Type	Description	Use Case
IPv4	Traditional IPv4 connectivity	Legacy internet access
IPv6	IPv6-only connectivity	IoT, modern networks
IPv4v6	Dual-stack IPv4 + IPv6	Default smartphone access
Ethernet	Layer-2 Ethernet frames	Industrial LAN extension, TSN
Unstructured	Raw IP or non-IP data	IoT sensors, point-to-point tunnels

N4 Interface and PFCP:

- **N4** connects SMF to UPF using **PFCP (Packet Forwarding Control Protocol)** — 3GPP TS 29.244
- PFCP is a UDP-based protocol (port 8805) with:
 - **Session Establishment:** Create forwarding context in UPF
 - **Session Modification:** Update rules (e.g., handover, QoS change)
 - **Session Deletion:** Remove forwarding context
 - **Heartbeat:** Detect UPF failure

Session Anchor:

- The **PDU Session Anchor (PSA)** is the UPF that maintains a stable point of attachment to the Data Network
- Provides IP address continuity during UE mobility
- Does NOT change during handover (unlike intermediate UPFs)
- SMF may insert Intermediate UPFs (I-UPF) for local routing while keeping PSA stable

3.3 User Plane Function (UPF)

The UPF is the **only user-plane NF** in the 5G Core. It handles all packet processing, forwarding, and policy enforcement for user data.

Key Responsibilities:

- **Packet Routing and Forwarding:** Routes packets between RAN and Data Network
- **Traffic Detection and Reporting:** Performs DPI for application identification
- **QoS Handling:** Enforces per-flow QoS (marking, policing, shaping)
- **Usage Reporting:** Measures traffic volume/duration for charging
- **Uplink Classifier (UL-CL):** Steers traffic to different data networks based on traffic matching
- **Branching Point:** Supports multi-homed PDU sessions
- **Buffering:** Buffers downlink packets for idle UEs, triggers DDN to SMF

Interfaces:

Interface	Connection	Protocol	Purpose
N3	gNB ↔ UPF	GTP-U over UDP	User plane tunnel from RAN
N6	UPF ↔ DN	IP / Ethernet	Exit point to Data Network
N9	UPF ↔ UPF	GTP-U over UDP	Inter-UPF forwarding (chaining)
N4	SMF ↔ UPF	PCFP over UDP	Control of UPF from SMF

PCFP Rules (programmed by SMF via N4):

Rule	Full Name	Purpose
PDR	Packet Detection Rule	Matches incoming packets (source interface, IP filters, GTP TEID, QFI)
FAR	Forwarding Action Rule	Defines what to do (forward, drop, buffer, duplicate); specifies destination
QER	QoS Enforcement Rule	Rate limiting (MBR/GBR), gating (open/close), reflective QoS
URR	Usage Reporting Rule	Volume/time/event-based measurement triggers for charging
BAR	Buffering Action Rule	Controls buffering behavior, notification parameters

Traffic Detection Methods:

- **SDF Filter:** 5-tuple IP filter (src/dst IP, src/dst port, protocol)
- **Application ID:** Pre-configured or dynamically learned via DPI
- **GTP-U TEID:** Identifies specific GTP tunnels
- **QFI (QoS Flow Identifier):** Maps to specific QoS flows

3.4 Unified Data Management (UDM)

The UDM provides subscriber data management, authentication credential handling, and subscription authorization. It is the 5G equivalent of the HSS.

Key Responsibilities:

- **SUPI Storage:** Stores and manages the Subscription Permanent Identifier (SUPI = IMSI or NAI format)
- **Authentication Credential Computation:** Generates authentication vectors (AV) using subscriber's long-term key (K) and operator parameters (OP/OPc)
- **Subscription Profile Management:** Maintains subscribed services, DNN authorizations, QoS profiles
- **Registration Management:** Tracks serving AMF/SMF per subscriber (NF registration)

- **DNN Authorization:** Validates whether a UE is authorized to access a specific DNN/slice combination
- **SUCI De-concealment:** Resolves SUCI back to SUPI (using SIDF — Subscription Identifier De-concealing Function)

Key Data Stored:

- SUPI, SUCI resolution keys
- Subscriber authentication credentials (K, OP/OPc, SQN)
- Subscribed S-NSSAI and DNN list
- Default session QoS parameters
- Serving NF registrations (AMF, SMF)
- Access and mobility subscription data

Interfaces:

- **Nudm:** Service-based interface exposed by UDM
- **N8 (via Nudm):** AMF retrieves subscription/mobility data
- **N10 (via Nudm):** SMF retrieves session management subscription data
- **N13 (via Nudm):** AUSF retrieves authentication credentials

Relationship to UDR:

- UDM contains application logic (auth vector computation, SUCI de-concealment)
- UDR is the pure data store backend
- UDM accesses UDR via **Nudr** interface

3.5 Authentication Server Function (AUSF)

The AUSF acts as the authentication server for the 5G network. It executes authentication procedures between the UE and the home network.

Key Responsibilities:

- **Authentication Execution:** Runs 5G-AKA or EAP-AKA' procedures
- **K_{AUSF} Derivation:** Derives the anchor key (K_{AUSF}) from which all subsequent keys are derived
- **Authentication Result Storage:** Stores authentication status for future reference
- **Home Network Authority:** Ensures authentication decision is made by the home network (even in roaming)

Authentication Methods:

Method	Description	Use Case
5G-AKA	5G Authentication and Key Agreement — enhanced AKA with home network confirmation	Primary method for USIM-based UEs
EAP-AKA'	Extensible Authentication Protocol with AKA prime — EAP framework-based	Roaming, non-3GPP access, IoT

Key Hierarchy from AUSF:

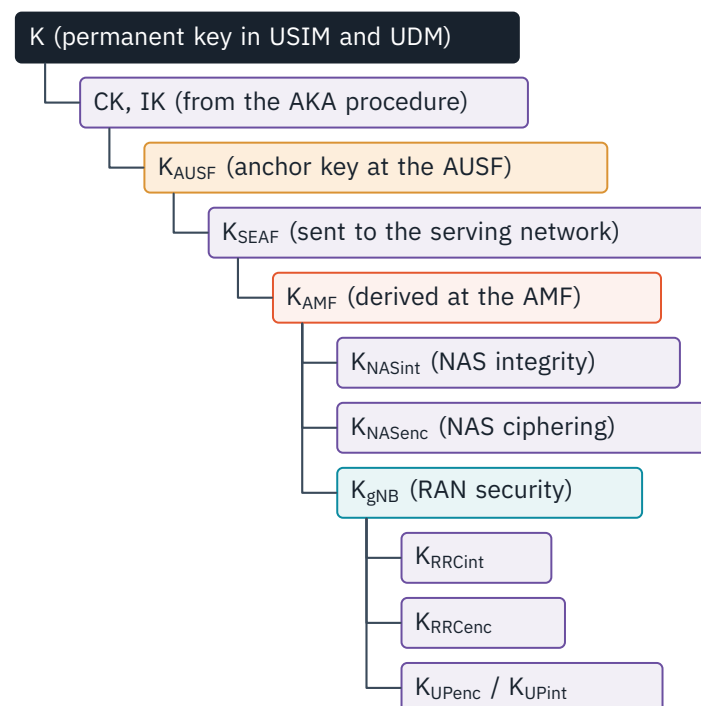


Figure 19.3 5G key hierarchy rooted at the AUSF

Interfaces:

- **Nausf:** Service-based interface exposed by AUSF
- **N12 (via Nausf):** AMF initiates authentication
- **N13 (via Nudm):** AUSF retrieves authentication vectors from UDM

3.6 Policy Control Function (PCF)

The PCF is the centralized policy engine of the 5GC. It provides policy rules to control network behavior for access, sessions, and UE-specific policies.

Key Responsibilities:

- **PCC Rule Generation:** Creates Policy and Charging Control rules for QoS and charging
- **AM Policy:** Access and Mobility policies (RFSP index, service area restrictions, RAT/frequency selection)

- **SM Policy:** Session Management policies (QoS, gating, charging, traffic steering)
- **UE Policy:** Policies delivered to the UE (URSP rules — UE Route Selection Policy, ANDSP)
- **Policy Decision:** Makes real-time policy decisions based on subscription, network conditions, AF requests

Policy Types:

Policy	Scope	Delivered To	Purpose
AM Policy	Access & Mobility	AMF (N15)	Service area, RFSP, RAT restriction
SM Policy	Session	SMF (N7)	QoS flows, PCC rules, charging
UE Policy	UE behavior	UE (via AMF)	URSP, ANDSP, WLANSF

PCC Rule Contents:

- Rule identifier and precedence
- Traffic flow template (SDF filter or application ID)
- QoS parameters (5QI, ARP, MBR, GBR)
- Gating status (open/close for UL/DL)
- Charging method (online/offline) and rating group
- Traffic steering policy

Interfaces:

- **Npcf:** Service-based interface exposed by PCF
- **N7 (via Npcf):** SMF retrieves SM policy decisions
- **N15 (via Npcf):** AMF retrieves AM policy decisions
- **N5 (via Npcf):** AF influences policy via NEF or directly

3.7 Network Slice Selection Function (NSSF)

The NSSF assists in selecting the appropriate network slice instance and AMF set for a UE.

Key Responsibilities:

- **Allowed NSSAI Determination:** Determines which S-NSSAIs the UE is permitted to use in the serving PLMN
- **S-NSSAI Mapping:** Maps Requested NSSAI to Configured NSSAI (home↔serving PLMN mapping in roaming)
- **AMF Redirection:** When the current AMF cannot serve the selected slice, NSSF identifies a target AMF set that can
- **NRF Selection:** Helps select appropriate NRF instance for slice-specific NF discovery

Key Concepts:

Term	Description
S-NSSAI	Single NSSAI = SST (8-bit Slice/Service Type) + optional SD (24-bit Slice Differentiator)
Requested NSSAI	S-NSSAIs sent by UE in Registration Request
Allowed NSSAI	S-NSSAIs permitted by network (returned in Registration Accept)
Configured NSSAI	S-NSSAIs provisioned in the UE by the HPLMN/VPLMN
Subscribed S-NSSAI	Slice subscription in UDM

Standard SST Values:

- SST=1: eMBB (enhanced Mobile Broadband)
- SST=2: URLLC (Ultra-Reliable Low-Latency Communications)
- SST=3: MIoT (Massive IoT)
- SST=4: V2X (Vehicle-to-Everything)

3.8 Network Repository Function (NRF)

The NRF is the central registry for all NF instances in the 5GC. It enables service discovery and authorization.

Key Responsibilities:

- **NF Registration:** NFs register their profiles (type, capacity, supported services, PLMN, slice, location)
- **NF Discovery:** Consumer NFs query NRF to find producer NFs matching criteria
- **NF Status Notification:** Notifies subscribed NFs about NF status changes (registered, deregistered, suspended)
- **OAuth2 Token Issuance:** Issues access tokens for service-based interface authorization

NF Profile Contents:

- NF Instance ID (UUID)
- NF Type (AMF, SMF, UPF, etc.)
- NF Status (REGISTERED, SUSPENDED, UNDISCOVERABLE)
- PLMN ID(s) served
- S-NSSAI(s) supported
- FQDN or IP addresses
- Supported services and versions
- Capacity, priority, load information
- Geographic locality

Discovery Query Parameters:

- Target NF Type
- Requester NF Type
- Service names
- Target PLMN, S-NSSAI, DNN
- Preferred locality

OAuth2 in 5GC:

- NRF acts as OAuth2 Authorization Server
- Consumer NF requests token with target NF type and services
- Token included in SBI requests (Authorization header)
- Producer NF validates token before serving the request

3.9 Network Exposure Function (NEF)

The NEF provides a secure framework for exposing 5GC capabilities and events to external Application Functions (AFs).

Key Responsibilities:

- **Exposure Framework:** Exposes network capabilities to 3rd-party AFs via standardized APIs
- **Traffic Influence:** AF can request specific traffic routing (e.g., route traffic to local edge server)
- **Monitoring Events:** AF subscribes to UE events (reachability, location, connectivity status)
- **Parameter Provisioning:** AF provisions expected UE behavior, background data transfer policies
- **Translation:** Maps between external identifiers (GPSI, external group ID) and internal identifiers (SUPI)
- **Security Gateway:** Prevents direct AF access to internal NFs

Key Exposure Capabilities:

Capability	Description
Monitoring Events	UE reachability, location reporting, loss of connectivity, roaming status
Traffic Influence	Request traffic routing to local Data Network (edge computing)
Analytics Exposure	Expose NWDAF analytics to external AFs
Parameter Provisioning	Expected UE behavior, background data transfer
Device Triggering	Trigger specific action on IoT device
NIDD	Non-IP Data Delivery for IoT

3.10 Unified Data Repository (UDR)

The UDR is the backend data store for subscriber, policy, and exposure data. It is accessed by UDM, PCF, and NEF.

Key Responsibilities:

- **Subscription Data Storage:** Stores subscriber profiles accessed by UDM
- **Policy Data Storage:** Stores policy data (PCC rules, AM/SM policies) accessed by PCF
- **Exposure Data Storage:** Stores exposure subscriptions and data accessed by NEF
- **Application Data Storage:** Stores AF-provided data (e.g., PFDs — Packet Flow Descriptions)

Data Categories:

Data Type	Consumer NF	Examples
Subscription Data	UDM	Authentication credentials, subscribed S-NSSAI, DNN configs
Policy Data	PCF	PCC rule templates, sponsored data policies
Exposure Data	NEF	Monitoring subscriptions, AF influence requests
Application Data	NEF/PCF	PFDs, background data transfer policies

Interface:

- **Nudr:** Service-based interface (the only interface of UDR)
- Accessed by UDM (Nudr_DataManagement), PCF (Nudr_DataManagement), NEF (Nudr_DataManagement)

3.11 Unstructured Data Storage Function (UDSF)

The UDSF enables stateless design of NFs by providing external storage for NF context data.

Key Responsibilities:

- **Unstructured Data Storage:** Stores arbitrary NF context/state as opaque data blocks
- **Stateless NF Support:** NFs can recover state after restart or load-balance across instances
- **CRUD Operations:** Create, Read, Update, Delete operations on stored records

Why UDSF Matters:

- Without UDSF, each NF instance holds session state in memory
- With UDSF, any NF instance can serve any request by retrieving state from UDSF
- Enables true horizontal scaling, failover, and cloud-native deployment
- Each NF type may have its own UDSF instance

Interface:

- **Nudsf:** Service-based interface for storing/retrieving unstructured data

3.12 Service Communication Proxy (SCP)

The SCP is a routing and load-balancing element placed in the SBI communication path between NFs. Defined in TS 23.501 Release 16+.

Key Responsibilities:

- **Indirect Communication:** NFs send requests to SCP instead of directly to the target NF
- **Load Balancing:** Distributes requests across multiple NF instances
- **Message Routing:** Routes messages based on discovery parameters without consumer NF needing full discovery
- **Topology Hiding:** Hides internal NF topology from external networks (roaming)
- **Overload Control:** Implements congestion handling and request throttling

Communication Models:

Model	Description
Model A (Direct)	NF consumer discovers target via NRF, communicates directly
Model B (Indirect w/o delegated discovery)	NF discovers target via NRF, routes via SCP
Model C (Indirect w/ delegated discovery)	NF sends to SCP, SCP performs discovery and routing
Model D (Indirect w/ delegated discovery + selection)	SCP handles everything: discovery, selection, routing

Benefits:

- Simplifies NF implementation (NFs don't need complex discovery/retry logic)
- Centralized observability and telemetry
- Easier deployment of cross-cutting concerns (mTLS termination, tracing)
- Enables seamless NF scaling without reconfiguration

4 Control Plane vs User Plane in 5G Core

The 5GC enforces a strict separation between Control Plane (CP) and User Plane (UP), known as CUPS (Control and User Plane Separation).

Control Plane (CP)

- **All NFs except UPF** operate in the control plane
- Communication via **Service-Based Interfaces (SBI)** using HTTP/2 + JSON
- Protocol stack: HTTP/2 → TCP → IP (or via SCP)
- Functions: signaling, policy, authentication, registration, session management, discovery
- Deployed centrally or regionally depending on latency requirements
- AMF and SMF are the primary CP anchors for UE signaling and session management

■ User Plane (UP)

- Only **UPF** operates in the user plane
- Programmed by SMF via **N4 / PFCP** (control interface)
- User data protocol stack: App Data → PDU Layer → GTP-U → UDP → IP
- Functions: packet forwarding, QoS enforcement, traffic detection, usage measurement, buffering
- Can be deployed at edge, regional, or central locations depending on latency and data locality needs
- Multiple UPFs can be chained (via N9) for service function chaining or local breakout

■ CP/UP Interaction Summary

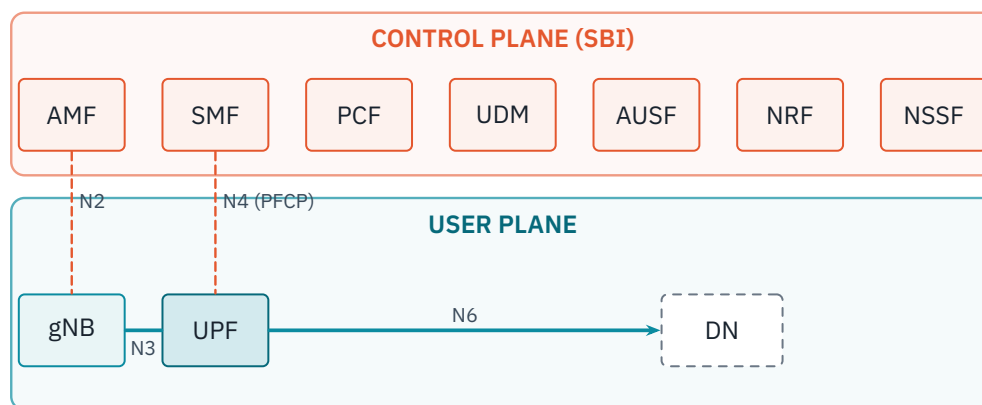


Figure 19.4 Control plane (SBI) and user plane of the 5G core

5 Major Comparison: 4G EPC vs 5G Core

Aspect	4G EPC	5G Core (5GC)
Architecture Style	Point-to-point interfaces between monolithic NEs	Service-Based Architecture (SBA), microservices
Signaling Protocol	GTP-C (v2), Diameter	HTTP/2 + JSON (RESTful APIs)
User Plane	S-GW + P-GW (combined or separated)	UPF (single user-plane NF)
Subscriber Database	HSS (Diameter-based)	UDM + UDR (SBI-based, separated logic/storage)
Authentication	EPS-AKA via HSS	5G-AKA / EAP-AKA' via AUSF + UDM
Policy	PCRF (Gx, Rx, Sd interfaces)	PCF (Npcf SBI, AM/SM/UE policies)
Session Management	MME + S-GW + P-GW combined	SMF (dedicated NF, decoupled from mobility)
Mobility Management	MME (combined with session signaling)	AMF (pure mobility/access, no session logic)
Network Slicing	Not natively supported	Native support (S-NSSAI, NSSF, per-slice NF instances)
Interface Style	Fixed reference points (S1, S5, S6a, S11...)	SBI services + reference points for UP
State Management	Stateful NEs	Stateless NFs + UDSF for external state
Deployment	Appliance-based, monolithic	Cloud-native, containerized, microservices
Discovery	Preconfigured or DNS-based	Dynamic via NRF (registration + discovery + OAuth2)
Edge Computing	Difficult (LIPA/SIPTO limited)	Native (distributed UPF, LADN, MEC integration)

4G to 5G Network Function Mapping

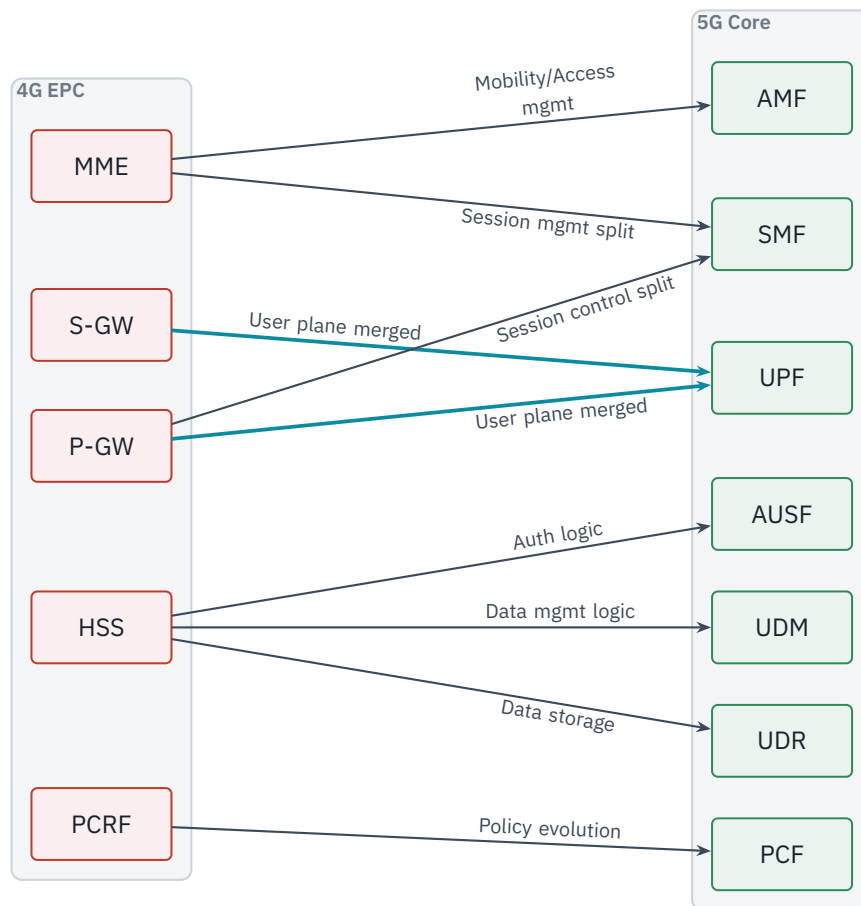


Figure 19.5 How the EPC functions map to 5G core network functions

6 Reference Points

Reference points define the interfaces between specific NF pairs. While SBI is used for CP-to-CP communication, reference points remain important for understanding protocol specifics.

Complete Reference Point Table

Reference Point	Connection	Protocol/Transport	Purpose
N1	UE ↔ AMF	NAS over RRC (via gNB)	NAS signaling (registration, session mgmt requests)
N2	RAN ↔ AMF	NGAP over SCTP	RAN-CN signaling (context setup, handover, paging)
N3	RAN ↔ UPF	GTP-U over UDP/IP	User plane tunnel carrying PDU session data
N4	SMF ↔ UPF	PCFP over UDP (port 8805)	UPF programming (PDR, FAR, QER, URR rules)
N6	UPF ↔ DN	IP / Ethernet	User traffic exit to Data Network
N7	SMF ↔ PCF	SBI (HTTP/2)	SM policy (PCC rules, QoS decisions)
N8	AMF ↔ UDM	SBI (HTTP/2)	Access/mobility subscription data
N9	UPF ↔ UPF	GTP-U over UDP/IP	Inter-UPF user plane forwarding
N10	SMF ↔ UDM	SBI (HTTP/2)	Session management subscription data
N11	AMF ↔ SMF	SBI (HTTP/2)	PDU session lifecycle management
N12	AMF ↔ AUSF	SBI (HTTP/2)	Authentication initiation
N13	AUSF ↔ UDM	SBI (HTTP/2)	Authentication vector retrieval
N14	AMF ↔ AMF	SBI (HTTP/2)	Inter-AMF mobility (UE context transfer)
N15	AMF ↔ PCF	SBI (HTTP/2)	AM policy (access/mobility restrictions)
N22	AMF ↔ NSSF	SBI (HTTP/2)	Slice selection assistance
N27	NRF ↔ NRF	SBI (HTTP/2)	Inter-PLMN NF discovery (via SEPP)
N32	SEPP ↔ SEPP	PRINS / TLS	Inter-PLMN security (roaming)

■ Key Reference Point Details

N1 (UE ↔ AMF):

- Carries NAS PDUs (Registration Request/Accept, Authentication, Security Mode, PDU Session Establishment)
- NAS messages are transparently relayed by gNB (encapsulated in NGAP on N2)
- Supports NAS security (integrity + ciphering) after security activation

N2 (RAN ↔ AMF):

- NGAP protocol over SCTP
- Procedures: Initial UE Message, PDU Session Resource Setup, Handover Required/Request, Paging
- Carries UE-associated and non-UE-associated signaling

N3 (RAN ↔ UPF):

- GTP-U tunnel identified by TEID (Tunnel Endpoint Identifier)
- One GTP-U tunnel per PDU session per gNB-UPF pair
- QoS Flow Identifier (QFI) carried in GTP-U extension header to map QoS flows

N4 (SMF ↔ UPF):

- PFCP (Packet Forwarding Control Protocol) — 3GPP TS 29.244
- Session-level management (Establishment, Modification, Deletion)
- Node-level management (Association Setup, Heartbeat)
- Reports: Usage Reports, Downlink Data Notification, End Marker

N9 (UPF ↔ UPF):

- GTP-U tunnel for multi-hop user plane paths
- Used in UL-CL (Uplink Classifier) and branching point scenarios
- Enables service function chaining and distributed UPF deployment

7 Deployment Options

The 5GC's modular architecture allows flexible deployment topologies based on service requirements.

7.1 Centralized Deployment

All NFs (including UPF) deployed in a central data center.

Characteristics:

- Simple operations, single site
- Higher latency for user plane (all traffic traverses central site)

- Suitable for wide-area coverage with relaxed latency requirements
- Lower infrastructure cost

7.2 Distributed Deployment

Control plane centralized, user plane distributed closer to users.

Characteristics:

- CP NFs (AMF, SMF, PCF, etc.) in regional/central DC
- UPF deployed at regional aggregation points
- Reduced user-plane latency
- SMF manages multiple UPF instances across locations

7.3 Edge UPF Deployment (MEC Integration)

UPF instances deployed at the network edge (cell site or local aggregation point) for ultra-low latency.

Characteristics:

- Local breakout of traffic at the edge
- Enables Multi-access Edge Computing (MEC) — ETSI MEC framework
- SMF can insert/remove edge UPFs dynamically based on UE location
- **UL-CL (Uplink Classifier):** Steers specific traffic flows to edge UPF while rest goes to central UPF
- **Local Area Data Network (LADN):** Data Network available only in specific geographic area
- Typical use cases: AR/VR, autonomous driving, industrial automation, cloud gaming

Deployment Architecture Comparison

Aspect	Centralized	Distributed	Edge UPF
CP Location	Central DC	Central/Regional DC	Central/Regional DC
UP Location	Central DC	Regional PoP	Cell site / Edge DC
User Plane Latency	High (10-50ms)	Medium (5-20ms)	Ultra-low (1-5ms)
Complexity	Low	Medium	High
Use Cases	General broadband	Regional services	URLLC, MEC, V2X
UPF Count	Few	Moderate	Many

■ PDU Session User Plane Path

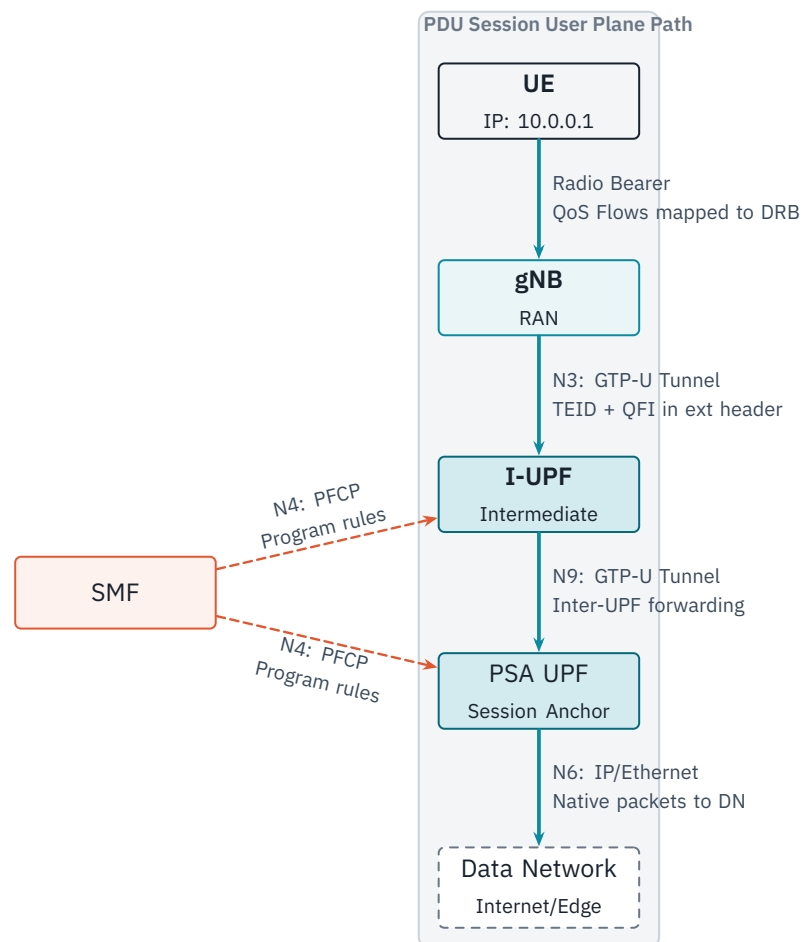


Figure 19.6 User-plane path of a PDU session with an intermediate UPF

PDU Session Establishment Flow (simplified):

1. UE → AMF: NAS PDU Session Establishment Request (via N1)
2. AMF → SMF: Nsmf_PDUSession_CreateSMContext (via N11)
3. SMF selects UPF, allocates IP address
4. SMF → UPF: PCF Session Establishment (via N4) — installs PDR/FAR/QER
5. SMF → AMF: N1N2MessageTransfer (contains NAS accept + N2 info)
6. AMF → gNB: PDU Session Resource Setup Request (via N2)
7. gNB → UPF: GTP-U tunnel established (N3)
8. UE receives PDU Session Establishment Accept with QoS rules

Summary

The 5G Core represents a paradigm shift from the 4G EPC:

- **SBA replaces point-to-point interfaces** — enabling flexibility, reusability, and independent NF evolution
- **Complete CUPS** — UPF is the only user-plane entity, programmable via PFCP
- **Cloud-native by design** — stateless NFs, UDSF, NRF-based discovery, containerization
- **Native slicing** — S-NSSAI, NSSF, per-slice NF selection from the ground up
- **Security enhanced** — SUCI concealment, home-network-controlled authentication, SBI OAuth2
- **Edge-ready** — distributed UPF, UL-CL, LADN for ultra-low-latency services

References: 3GPP TS 23.501 (System Architecture), TS 23.502 (Procedures), TS 29.244 (PFCP), TS 33.501 (Security Architecture)